Christopher Collier

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EDUCATION

University of Delaware
B.S. in Biomedical Engineering, Minor in Biomechanical Engineering; Incoming M.S. in Robotics, Fall 2027. Dean's List (6/6 semesters); Presidential Scholarship Recipient; McCauley Miriam Engineering Scholarship. GPA: 3.83/4.00. *May 2027*

EXPERIENCE

Engineering Research Intern May 2026 – August 2026
Nemours Children's Health (BRITE Lab)
- Redesigned microfluidic oxygenator sealing and validated 80 cN·m torque specification across 125–175 µm channel heights
- Characterized failure modes by identifying leaks and defects via dye perfusion, informing iterative design refinement
- Developed automated UV adhesive dispensing process for hollow fiber membrane bonding

Teaching Assistant January 2026 – May 2026
University of Delaware, Medical Device Design
- Guided 15 student teams through the full medical device design process
- Led SolidWorks (CAD) instruction and provided feedback on technical documentation

Teaching Assistant August 2026 – Present
University of Delaware, Senior Capstone
- Instruct Mechanical and Biomedical Engineering students in safe use of power and hand tools

Resident Assistant August 2024 – Present
University of Delaware, Residence Life and Housing
- Support and guide 40 students, resolving resident issues and responding to emergencies

PROJECTS

Dynamic Stability Footwear Tester August 2026 – Present
Mechanical/Biomedical Engineering Senior Capstone, Under Armour
- Design and prototype a footwear stability test fixture for an Instron ElectroPuls E10000 to evaluate inversion/eversion rotational stiffness under controlled vertical loading
- Develop SolidWorks models of custom adapters, foot-mold mounts, and bevel-gear mechanisms to convert actuator rotation into motion about the heel-to-toe axis
- Troubleshoot prototype load transfer, shaft coupling, and gearbox restraint while collaborating with industry sponsors and faculty to refine test requirements and fixture design

Radial Arterial Catheterization Wrist Brace February 2026 – May 2026
Biomedical Engineering Junior Design
- Designed a clinician-sponsored wrist brace locking at 0, 10, and 20 degrees of extension while preserving catheter access
- Built 3D-printed prototype in SolidWorks and authored a complete engineering drawing packet
- Validated structural compliance: sustained 7.5 lbs over one hour with no deformation at all three locked extension angles

Three-Compartment Pharmacokinetic Model of Propofol February 2026 – May 2026
Biomedical Modeling and Simulation
- Implemented a three-compartment PK model in MATLAB (ode15s) to simulate plasma and tissue drug concentrations
- Performed observability and controllability analysis on the state-space system, confirming full rank

Biomechanical Analysis of Squat Variations February 2026 – May 2026
Clinical Biomechanics
- Collected 3D kinematic data from 19 participants across 3 squat variations using OpenCap motion capture
- Identified significant differences in knee angle and movement velocity by squat type, informing rehabilitation loading recommendations

ACTIVITIES & LEADERSHIP

Delaware Men's Club Volleyball September 2023 – Present
Captain and Vice President
- Rookie of the Year (2024), A Team MVP (2025, 2026), Defensive Player of the Year (2025, 2026)

SKILLS

CAD/Design: SolidWorks, Onshape, Autodesk Fusion
Fabrication: 3D printing (Bambu Lab, Formlabs), laser cutting, automated dispensing, soldering
Analysis/Programming: MATLAB, Python, JMP, LTSpice, Waveforms, Arduino
Software: Microsoft 365, Google Workspace